

Professional Summary

Results-driven Senior Data Engineer with over 5+ years of experience designing, building, and optimizing enterprise-scale data platforms across E-commerce, Financial Services, and Banking domains. Proven expertise in developing high-performance ETL/ELT pipelines, real-time streaming architectures, and cloud-native data solutions on AWS and GCP, processing terabytes of data daily. Highly skilled in Python, PySpark, Apache Spark, Kafka, and modern data stack technologies, with a strong track record of reducing infrastructure costs by over 40% while improving system performance and reliability. Adept at architecting data lakes and lakehouse solutions, implementing data governance frameworks, and enabling analytics platforms that support data-driven decision-making for millions of users.

Technical Skills

- **Programming:** Python, PySpark, SQL, Scala, Java, Shell Scripting, R
- **Big Data:** Apache Spark, Kafka, Airflow, Hadoop, Hive, Presto, Flink, Beam
- **ETL/ELT:** Airflow, dbt, Informatica, Talend, AWS Glue, Azure Data Factory, NiFi, Fivetran
- **Cloud Platforms:** AWS (S3, Redshift, EMR, Glue, Lambda, Kinesis, Athena), GCP (BigQuery, Dataflow, Pub/Sub, Dataproc)
- **Databases:** PostgreSQL, MySQL, Oracle, SQL Server, MongoDB, Cassandra, DynamoDB, Redis, Elasticsearch
- **DevOps & Tools:** Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform, Ansible, Git

Professional Experience

eBay , Infobhan, emp: Infolance
Data Engineer

June 2023 – Present Senior
San Jose, CA

- Architected and implemented large-scale data pipelines supporting eBay marketplace serving 182 million active buyers and 19 million sellers globally, processing 1.7 billion annual transactions and 5 billion daily events using PySpark, Apache Kafka, and **AWS** infrastructure for analytics and ML applications.
- Built real-time streaming data platform using Apache Kafka, Flink, and AWS Kinesis processing 100 million+ events daily from product listings, bidding activities, purchases, and user interactions, enabling real-time personalization, fraud detection, and inventory management with sub-second latency.
- Designed data lakehouse architecture on AWS S3 and Snowflake consolidating 50+ petabytes of marketplace data including transactions, user behavior, product catalog, search logs, and seller metrics, enabling unified analytics for 3,000+ data scientists and analysts across organization.

Developed automated ETL pipelines using Apache Airflow, AWS Glue, and Python processing 10TB+ daily data ingestion from 300+ upstream sources including payment systems, shipping providers, customer service platforms, and third-party integrations ensuring 99.9% data freshness SLA.

- Implemented machine learning feature engineering pipelines using PySpark and Python generating 5,000+ real-time features for recommendation systems, search ranking, fraud detection, and dynamic pricing models, improving model performance by 25% and reducing feature computation latency by 60%.
 - Built seller analytics data platform using Spark and BigQuery analyzing performance metrics, listing optimization, pricing trends, and buyer engagement for 19M sellers, providing insights that increased average seller revenue by 22% through data-driven recommendations.
 - Optimized Snowflake data warehouse performance through clustering keys, materialized views, and query optimization reducing query execution time by 70% and lowering compute costs by 45% while maintaining subsecond response times for business intelligence dashboards.
 - Implemented data quality framework using Great Expectations and dbt tests monitoring 10,000+ data pipelines and validating 1 billion+ records daily, reducing data quality incidents by 65% and improving trust in analytics through automated anomaly detection and alerting.
 - Designed fraud detection data pipelines using Kafka and Spark Streaming processing transaction data in real-time, generating fraud signals and risk scores that prevented \$35M in fraudulent transactions annually with 92% precision and 88% recall.
 - Built search and recommendation data infrastructure using Elasticsearch and Redis processing 200 million+ search queries daily, powering personalized product recommendations that increased conversion rates by 28% and average order value by 18%.
 - Established CI/CD pipelines for data workflows using Jenkins, GitLab, and Terraform enabling automated testing, deployment, and infrastructure provisioning, reducing deployment time from days to hours and achieving 99.7% pipeline reliability.
 - Mentored team of 3 junior data engineers on Spark optimization, data modeling, and cloud best practices, conducting code reviews and establishing data engineering standards improving team productivity by 30%
- Data Engineer II July 2021 – Aug 2024

Developed and maintained data pipelines supporting Fidelity's wealth management and retirement services managing \$4.5+ trillion in assets for 43+ million customers, processing financial transactions, portfolio holdings, market data, and customer interactions using Informatica, Oracle, and AWS.

Built ETL workflows using Informatica PowerCenter, AWS Glue, and Python processing 200+ million daily transactions from trading platforms, mutual fund accounting, 401(k) recordkeeping, and brokerage systems ensuring data accuracy for portfolio valuation and performance reporting.

Environment: Python, PySpark, SQL, Informatica PowerCenter, Apache Spark, Apache Kafka, Apache Airflow, AWS (S3, EMR, Glue, Redshift, Kinesis, Lambda), Snowflake, Teradata, Oracle, PostgreSQL, MongoDB, Docker, Jenkins, Git, ETL, Financial Data, Investment Management, Regulatory Compliance

Fidelity Investments, Hyderabad, India , Capegemini, Indus Digatech PVt Ltd

July 2021 – Aug 2022

Data Engineer II

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Designed dimensional data warehouse on Snowflake and Teradata consolidating financial data from 100+ source systems including trading engines, custody platforms, CRM, and market data vendors, enabling enterprise analytics and regulatory reporting for investment operations.

- Implemented real-time market data ingestion using Apache Kafka and Spark Streaming processing 10 million+ price updates per minute from stock exchanges, mutual fund NAV systems, and bond pricing services, enabling accurate portfolio valuation and risk analysis with sub-second latency.
- Developed customer analytics data platform using PySpark and AWS Redshift analyzing investment behavior, product holdings, and digital engagement for 43M customers, enabling personalized financial advice and improving retention rates by 18% through targeted communications.
- Built regulatory reporting pipelines using Informatica and SQL generating automated SEC filings, FINRA reports, and Form 5500 retirement plan disclosures processing millions of transactions ensuring 100% compliance with regulatory deadlines and audit requirements.
- Implemented data quality validation framework using Python and SQL monitoring 500M+ daily records for completeness, accuracy, and consistency across trading, accounting, and customer data, reducing reconciliation breaks by 75% and improving data reliability.
- Designed 401(k) retirement plan data pipelines using Spark and Airflow processing participant contributions, employer matches, investment elections, and loan/distribution transactions for 23 million retirement accounts, ensuring accurate recordkeeping and compliance.
- Optimized batch processing workflows reducing overnight data processing time from 10 hours to 5 hours through parallel processing, incremental loading, and database tuning, ensuring timely availability of portfolio data for morning trading and customer service.
- Built data reconciliation processes using Python and SQL comparing trading data, custody records, and fund accounting systems identifying discrepancies, automating 85% of reconciliation tasks saving 400+ hours monthly and reducing settlement breaks.
- Developed investment performance attribution data pipelines using Spark analyzing portfolio returns, benchmark comparisons, and factor exposures for 10,000+ mutual funds and separately managed accounts, supporting performance reporting and investment analysis.
- Collaborated with quantitative analysts and portfolio managers to build data infrastructure supporting factor analysis, risk models, and algorithmic trading strategies, processing 30+ years of historical market data for back testing and research.

Environment: Python, PySpark, SQL, Informatica PowerCenter, Apache Spark, Apache Kafka, Apache Airflow, AWS (S3, EMR, Glue, Redshift, Kinesis, Lambda), Snowflake, Teradata, Oracle, PostgreSQL, MongoDB, Docker, Jenkins, Git, ETL, Financial Data, Investment Management, Regulatory Compliance

**Discover Financial Services,
Data Engineer**

**Hyderabad, India
Jan 2020 – July 2021**

- Designed and implemented data pipelines supporting Discover's credit card operations serving 60+ million cardmembers and processing \$150+ billion in annual purchase volume, building ETL workflows using Informatica, Teradata, and Hadoop for analytics and risk management applications.

- Built credit card transaction processing data platform using Hadoop, Hive, and Spark analyzing 3 billion+ annual transactions, generating insights for fraud detection, credit risk assessment, and marketing campaigns that improved response rates by 35% and reduced fraud losses by \$15M.
- Developed fraud detection data pipelines using Kafka and Spark Streaming processing real-time authorization requests and transaction data, generating fraud scores and alerts that prevented \$25M in fraudulent charges annually with 90% detection accuracy and minimal false positives.

Implemented customer analytics data warehouse on Teradata and Snowflake consolidating cardholder demographics, spending patterns, payment behavior, and customer service interactions for 60M customers, enabling targeted marketing and retention strategies improving lifetime value by 20%.

Built credit risk analytics pipelines using Python and SQL processing credit bureau data, payment history, and utilization patterns for \$90B+ credit portfolio, supporting underwriting decisions and loss forecasting models that reduced default rates by 12%. • Designed ETL workflows using Informatica and Python integrating data from authorization systems, settlement networks, customer databases, and third-party vendors processing 10M+ daily records, ensuring data consistency for accounting, reporting, and compliance.

- Developed rewards program analytics using Spark analyzing cashback redemptions, point accruals, and program engagement for Discover's cashback rewards serving 60M cardmembers, optimizing program costs and improving member satisfaction scores by 18%.
- Implemented data quality monitoring using Python and SQL validating transaction data, customer information, and merchant details across 500+ data tables, reducing data errors by 60% and improving reliability of downstream analytics and machine learning models.
- Built regulatory compliance reporting pipelines using SQL and Informatica generating automated CFPB submissions, OCC reports, and Basel III capital calculations ensuring 100% on-time delivery and accuracy for federal regulatory requirements.
- Optimized Teradata data warehouse performance through indexing strategies, statistics collection, and query tuning reducing report execution time by 55% during month-end financial close and improving user productivity across finance and operations teams.
- Developed merchant analytics platform using Hadoop and Hive analyzing transaction patterns, authorization rates, and settlement data for 10 million+ merchants, identifying high-risk merchants and supporting network security initiatives.
- Automated monthly reconciliation processes using Python scripts comparing card network settlements, internal ledgers, and bank statements, reducing manual effort by 70% and ensuring accurate financial reporting for \$150B+ annual transaction volume.

Environment: Python, SQL, Informatica PowerCenter, Apache Spark, Apache Kafka, Hadoop, Hive, Teradata, Snowflake, Oracle, PostgreSQL, Unix/Linux, Shell Scripting, ETL, Credit Card Processing, Fraud Detection, Risk Analytics, Regulatory Compliance.

Education

**Saint Francis College,
Master of Science in Computer Science**

Brooklyn, NY
May 2024

□ Relevant Coursework: Big Data Analytics, Cloud Computing, Database Systems, Machine Learning, Advanced Algorithms, Distributed Systems